

```
In [1]: import matplotlib.pyplot as plt
        from sklearn.datasets import load_digits
        from sklearn.cluster import KMeans
        from sklearn.decomposition import PCA
```

1. Load datasets

```
In [2]: # Load data
        digits = load_digits()
```

```
In [3]: X = digits.data
        y = digits.target
```

2. Run K-Means

```
In [4]: kmeans = KMeans(n_clusters=10, init="k-means++", n_init=10, random_state=0,
        kmeans.fit(X)
```

Initialization complete
Iteration 0, inertia 1883216.0.
Iteration 1, inertia 1213908.616976922.
Iteration 2, inertia 1197396.240353104.
Iteration 3, inertia 1190134.7879864173.
Iteration 4, inertia 1184659.2111132473.
Iteration 5, inertia 1182293.3096599805.
Iteration 6, inertia 1181306.468904349.
Iteration 7, inertia 1180608.6287572219.
Iteration 8, inertia 1180078.019384263.
Iteration 9, inertia 1179467.517888174.
Iteration 10, inertia 1178028.9382105921.
Iteration 11, inertia 1175180.9398219474.
Iteration 12, inertia 1173115.396449361.
Iteration 13, inertia 1172172.602442375.
Iteration 14, inertia 1171676.4542999344.
Iteration 15, inertia 1171454.044960718.
Iteration 16, inertia 1171372.9194104664.
Iteration 17, inertia 1171302.3768490478.
Iteration 18, inertia 1171289.2204911562.
Converged at iteration 18: strict convergence.

Initialization complete
Iteration 0, inertia 1860851.0.
Iteration 1, inertia 1242404.3365244414.
Iteration 2, inertia 1219843.7377164494.
Iteration 3, inertia 1204580.6106078173.
Iteration 4, inertia 1191069.5191568949.
Iteration 5, inertia 1179337.6288021735.
Iteration 6, inertia 1172816.1561530607.
Iteration 7, inertia 1171173.4817817064.
Iteration 8, inertia 1170359.0974490384.
Iteration 9, inertia 1169732.6795866862.
Iteration 10, inertia 1169586.5570967891.
Converged at iteration 10: strict convergence.

Initialization complete
Iteration 0, inertia 2025300.0.
Iteration 1, inertia 1258928.596244437.
Iteration 2, inertia 1218239.0614200816.
Iteration 3, inertia 1204628.6305191726.
Iteration 4, inertia 1193371.6225511692.
Iteration 5, inertia 1186045.4542049367.
Iteration 6, inertia 1180329.418380258.
Iteration 7, inertia 1177801.5029243305.
Iteration 8, inertia 1176916.430928636.
Iteration 9, inertia 1176136.5971587596.
Iteration 10, inertia 1174749.508279437.
Iteration 11, inertia 1172643.6467617278.
Iteration 12, inertia 1171332.8528923895.
Iteration 13, inertia 1170250.287335398.
Iteration 14, inertia 1169683.9605646639.
Iteration 15, inertia 1169301.5441770616.
Iteration 16, inertia 1169160.4732742033.
Iteration 17, inertia 1169103.3333268226.
Iteration 18, inertia 1169069.5230511124.
Converged at iteration 18: strict convergence.

Initialization complete

Iteration 0, inertia 1971665.0.
Iteration 1, inertia 1256009.0566967018.
Iteration 2, inertia 1221260.0534744659.
Iteration 3, inertia 1202330.0476277706.
Iteration 4, inertia 1185303.7795089283.
Iteration 5, inertia 1178037.829090152.
Iteration 6, inertia 1175592.9474280393.
Iteration 7, inertia 1174763.2627549632.
Iteration 8, inertia 1174251.669747094.
Iteration 9, inertia 1173829.5090753643.
Iteration 10, inertia 1173423.211987007.
Iteration 11, inertia 1172544.200485769.
Iteration 12, inertia 1171386.4824950995.
Iteration 13, inertia 1170695.5106087928.
Iteration 14, inertia 1170422.1226989205.
Iteration 15, inertia 1170331.144731729.
Iteration 16, inertia 1170223.867344201.
Iteration 17, inertia 1170051.0048869208.
Iteration 18, inertia 1169933.0930959964.
Iteration 19, inertia 1169797.191919975.
Iteration 20, inertia 1169600.8693583559.
Iteration 21, inertia 1169539.282309758.
Iteration 22, inertia 1169476.761342185.
Iteration 23, inertia 1169464.3425675835.
Converged at iteration 23: strict convergence.

Initialization complete

Iteration 0, inertia 2082951.0.
Iteration 1, inertia 1273060.877565175.
Iteration 2, inertia 1216577.3782052156.
Iteration 3, inertia 1205218.748546253.
Iteration 4, inertia 1194126.696555729.
Iteration 5, inertia 1179678.2357348397.
Iteration 6, inertia 1168493.815867722.
Iteration 7, inertia 1166509.9960113715.
Iteration 8, inertia 1166017.6581042353.
Iteration 9, inertia 1165715.8259448898.
Iteration 10, inertia 1165463.925027714.
Iteration 11, inertia 1165410.4389941753.
Iteration 12, inertia 1165323.935417499.
Iteration 13, inertia 1165241.7330264712.
Iteration 14, inertia 1165212.93430505.
Iteration 15, inertia 1165198.7229303648.
Iteration 16, inertia 1165188.890449232.
Converged at iteration 16: strict convergence.

Initialization complete

Iteration 0, inertia 2009665.0.
Iteration 1, inertia 1314410.692991655.
Iteration 2, inertia 1242537.445494285.
Iteration 3, inertia 1204177.4877245282.
Iteration 4, inertia 1187480.2585421544.
Iteration 5, inertia 1179434.7774511091.
Iteration 6, inertia 1174698.4351444584.
Iteration 7, inertia 1171607.1003617172.
Iteration 8, inertia 1170082.918650638.
Iteration 9, inertia 1169693.9819403177.
Iteration 10, inertia 1169567.006181869.

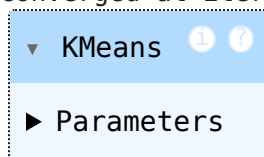
Iteration 11, inertia 1169529.1458540363.
Iteration 12, inertia 1169518.274344399.
Converged at iteration 12: strict convergence.
Initialization complete
Iteration 0, inertia 2057486.0.
Iteration 1, inertia 1268709.3426805916.
Iteration 2, inertia 1214505.8023898606.
Iteration 3, inertia 1190922.041433137.
Iteration 4, inertia 1174481.7865634626.
Iteration 5, inertia 1171837.209313846.
Iteration 6, inertia 1170933.7922556384.
Iteration 7, inertia 1170554.9910831323.
Iteration 8, inertia 1170415.330806748.
Iteration 9, inertia 1170295.1465913905.
Iteration 10, inertia 1170173.4611385283.
Iteration 11, inertia 1169993.3444724635.
Iteration 12, inertia 1169751.7755697125.
Iteration 13, inertia 1169659.9486026717.
Iteration 14, inertia 1169575.8790727435.
Iteration 15, inertia 1169551.519201519.
Converged at iteration 15: strict convergence.
Initialization complete
Iteration 0, inertia 2007469.0.
Iteration 1, inertia 1267000.0729362953.
Iteration 2, inertia 1232828.272340189.
Iteration 3, inertia 1223384.4174873927.
Iteration 4, inertia 1208733.3502733258.
Iteration 5, inertia 1181070.0699666082.
Iteration 6, inertia 1172262.9359133975.
Iteration 7, inertia 1171407.5311103568.
Iteration 8, inertia 1171249.6365349144.
Iteration 9, inertia 1171038.7674833762.
Iteration 10, inertia 1170824.2617183737.
Iteration 11, inertia 1170454.3000453617.
Iteration 12, inertia 1170268.796764465.
Iteration 13, inertia 1170204.6111563644.
Iteration 14, inertia 1170154.742894593.
Iteration 15, inertia 1170131.0756905566.
Iteration 16, inertia 1170100.796991956.
Iteration 17, inertia 1170077.976814122.
Converged at iteration 17: strict convergence.
Initialization complete
Iteration 0, inertia 1890117.0.
Iteration 1, inertia 1264433.8959547093.
Iteration 2, inertia 1236475.9404398773.
Iteration 3, inertia 1227166.2773051818.
Iteration 4, inertia 1223851.4351736451.
Iteration 5, inertia 1222481.0750488497.
Iteration 6, inertia 1221658.1983375847.
Iteration 7, inertia 1221287.9022501414.
Iteration 8, inertia 1220696.628890057.
Iteration 9, inertia 1220378.9360045244.
Iteration 10, inertia 1220148.7625193086.
Iteration 11, inertia 1220049.1721153061.
Iteration 12, inertia 1219779.6536051417.
Iteration 13, inertia 1219623.5491766108.

```

Iteration 14, inertia 1219459.8614223877.
Iteration 15, inertia 1219294.7375659458.
Iteration 16, inertia 1219080.2020493494.
Iteration 17, inertia 1218464.217448919.
Iteration 18, inertia 1217306.7172852573.
Iteration 19, inertia 1215346.5350699227.
Iteration 20, inertia 1214777.6799542431.
Iteration 21, inertia 1214614.7732662694.
Iteration 22, inertia 1214564.077023427.
Iteration 23, inertia 1214557.4026873934.
Converged at iteration 23: strict convergence.
Initialization complete
Iteration 0, inertia 1994764.0.
Iteration 1, inertia 1245235.7391268814.
Iteration 2, inertia 1201068.0813556921.
Iteration 3, inertia 1184145.5980600135.
Iteration 4, inertia 1172928.2699477514.
Iteration 5, inertia 1167207.3477777313.
Iteration 6, inertia 1165978.3266579134.
Iteration 7, inertia 1165635.4067334298.
Iteration 8, inertia 1165325.2628621734.
Iteration 9, inertia 1165270.841436491.
Iteration 10, inertia 1165237.384598685.
Iteration 11, inertia 1165228.0007319502.
Converged at iteration 11: strict convergence.

```

Out[4]:



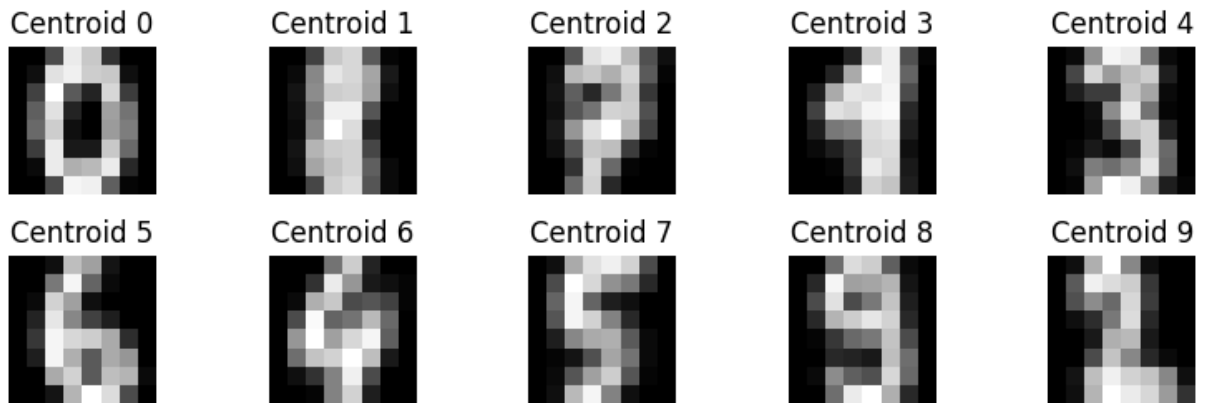
3. Visualize

```

In [5]: fig, axs = plt.subplots(2, 5, figsize=(8, 3))
        for i, ax in enumerate(axs.flat):
            ax.imshow(kmeans.cluster_centers_[i].reshape(8, 8), cmap='gray')
            ax.set_title(f'Centroid {i}')
            ax.axis('off')
        plt.suptitle('Cluster Centroids (K-Means)')
        plt.tight_layout()
        plt.show()

```

Cluster Centroids (K-Means)



4. Movement and separation in 2D with PCA

This shows how data and centroids look in low dimensions

```
In [6]: X_pca = PCA(n_components=2).fit_transform(X)
kmeans_pca = KMeans(n_clusters=10, init="k-means++", n_init=10, random_state=0)
labels_pca = kmeans_pca.fit_predict(X_pca)
centroids_pca = kmeans_pca.cluster_centers_

plt.figure(figsize=(8, 6))
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=labels_pca, s=10, cmap='tab10')
plt.scatter(centroids_pca[:, 0], centroids_pca[:, 1], marker='x', s=200, color='red')
plt.title("K-Means Clusters and Centroids (PCA-reduced digits)")
plt.show()
```

K-Means Clusters and Centroids (PCA-reduced digits)

